Chengqi Xue



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Robotics & Machine Learning & Computer Vision & Automation
Website: Chengqi Xue — Robotics and embodied AI

Education

MSc. Robotics | King's College London, UK **09/2025-09/2026**
Core Modules: *Robot Dynamics and Control, Robot Kinematics and Motion Planning, Sensing and Perception, Machine Learning, Intelligence and Autonomy*

BEng. Automation | Yangtze University, China **09/2020-07/2024**
GPA: 3.71/5.0 Ranking: Top 7% National Scholarship (4%)
Core Modules: *C Language Programming, Computer Control Technology, Analog Electronic Technology, Digital Electronic Technology, Principle of Automatic Control, Motion Control System, Machine Vision*

Research & Project Experience

Vision–Tactile–Language–Action Learning for Robotic Cloth Sorting | KCL 09/2025-08/2026

- Proposed a VTLA framework extending Vision–Language–Action (VLA) models with a tactile modality for contact-rich manipulation tasks
- Designed a multi-modal tokenization pipeline fusing vision, tactile, and language inputs, fine-tuning Qwen2.5-VL-7B with QLoRA on 3,937 fabric-comparison conversations
- Developed a tactile-image (GelSight sensing) force estimator for finray grippers to improve grasp stability during cloth sorting
- Developing a π0.5VLA-based language-conditioned action policy for Franka cloth manipulation

Tactile UMI: Hand-held Visuotactile Demonstrations to a π0.5 Policy| KCL 09/2025-08/2026

- Built a UMI-style hand-held demonstration gripper extended with dual GelSight fingertips, a wrist fisheye camera, Meta Quest 6-DoF tracking and ArUco-based gripper-width sensing.
- Turned four sensor streams into one robot-ready timeline: hand–eye calibration from 40 pose pairs, measured −56.7 ms Quest–camera and +25 ms tactile–RGB offsets by motion alignment, a five-point gripper-width lookup, and per-episode audit tools;
- Fine-tuned π0.5 on the corpus converted to LeRobot format, served it through a websocket inference server and deployed it autonomously on a Franka FR3

Socially assistive TIAGo Robot for Chemotherapy Wards | KCL 09/2025-04/2026

- Built a TIAGo robot simulation in Webots integrated with ROS 2, RViz, SLAM, and Navigation2 for autonomous indoor assistive robot research
- Developed a full SLAM and navigation pipeline enabling environment mapping, localization, global/local path planning, and obstacle avoidance
- Designed a perception module combining speech recognition and waving detection to identify patients needing assistance and supply navigation targets
- Implemented AMCL-based pose estimation to improve reliable navigation in structured indoor environments

Research Assistant 03/2022-07/2025
Multi-Vision and Deep-Learning Inspection of Internal Threads,Yangtze University

- **Published 2 journal papers, 2 conference papers, and 2 patents**
- Designed a six-fisheye ring probe (two rings of three cameras at 120°, LED ring light) with Zhang calibration and fisheye undistortion for dark, curved thread bores.
- Proposed cylindrical-projection panorama stitching (SIFT + RANSAC + Laplacian-pyramid blending) that unrolls curved thread surfaces where planar stitching fails.
- Tackled defect scarcity with DFMGAN few-shot defect synthesis; YOLOv8 reached mAP 91.6% (SSD 71.1%), and a robot-arm carrier cut acquisition time by 44%.

Research Assistant 03/2021- 07/2025

Small-Sample Machine Learning for Engineering-Structure Modelling and Design,Yangtze University

- **Published 2 journal papers**
- With only 72 destructive tests, designed genetic-algorithm data augmentation (72 → 15,000 virtual samples) and a Reptile meta-learning regressor: R^2 0.982 for ultimate bearing capacity and 0.955 for displacement, versus ≤ 0.875 for SVR/GPR/RBFNN baselines.
- Benchmarked TPOT AutoML against classical learners on two GFRP column datasets (best mean R^2 0.996) and inverted the model into optimal section designs

Research Assistant 06/2023-02/2024

Industrial Monitoring and Early Warning System Based on Machine Vision, Yangtze University

- **2 granted utility patents**
- Designed a 3-DoF wellhead camera gantry that flags missing hard hats and smoking
- Designed a solar-powered H_2S alarm beacon coupled to a wind vane that points workers to the up-wind escape route

Publications

1. T. Deng,**C. Xue (Co-first Author)**, G. Zhang, "*Data Modeling Analysis of GFRP Tubular Filled Concrete Column Based on Small Sample Deep Meta Learning Method*," *PLOS ONE* https://doi.org/10.1371/journal.pone.0305038
2. X. Dou, **C. Xue (Co-first Author)**, G. Zhang, Z. Jiang, "*Internal Thread Defect Detection System Based on Multi-vision,*" *PLOS ONE*. https://doi.org/10.1371/journal.pone.0304224
3. P. Shu, **C. Xue (Co-first Author)**,G. Zhang, T. Deng, *"Study on Design Optimization of GFRP Tubular Column Composite Structure Based on Machine Learning Method". PLOS ONE.* https://doi.org/10.1371/journal.pone.0301865
4. Z. Jiang, X. Dou, X. Liu, **C. Xue** , et al. *"Internal Thread Defect Generation Algorithm and Detection System Based on Generative Adversarial Networks and You Only Look Once". Sensors* 2024, 24(17),5636. https://doi.org/10.3390/s24175636
5. H. Yuan, X. Dou, **C. Xue (Corresponding Author)**, et al. "Review of image calibration and distortion correction based on internal threads" Proc. SPIE (AASIP 2024) https://doi.org/10.1117/12.3045519
6. Z. Xu, **C. Xue (Corresponding Author)**, X. Dou, et al., "A Review of Image Stitching Algorithms Based on *Internal Thread". Proc. SPIE* (RSTIP 2024). http://dx.doi.org/10.1117/12.3065383
7. Y. Zhou, C. Xue, M. Li, et al., "From Weak Cues to Visual Confirmation: Cross-View Hypothesis Memory for Active Target Search Under Occlusion," IEEE Access (Submitted, 2026).

Patents

First Inventor, **A Wellhead Monitoring Device Based on Machine Vision**, Patent No.: ZL 202323487863.X

First Inventor, **A Wind Event Early Warning System**, Patent No.: ZL 2023 2 3530543.8

Third Inventor, **A Device for Inspecting External Threads of Tubular Tools**, Patent No.: 202411050637.1

Third Inventor, **Internal Thread Defect Detection System**, Patent No.: 202411066290.X

Honors & Awards

Best Researcher Award at the International Research Awards on Sensing Technology	09/2024
National Scholarship, Yangtze University	12/2023
Third prize, (Provincial level), National Undergraduate Electronic Design Contest	10/2023
First Prize, (National level), College students AI Technology Competition Knowledge Contest	09/2023
Second Prize, College Students "Internet" Innovation and Entrepreneurship Competition	07/2022

Skills

Proficient in using Python, MATLAB, Linux, RViz, ROS 2, SLAM, Navigation2 Gazebo, and etc.